

A MANAGER 360°: COMMUNICATION ARTS AND MANAGEMENT INNOVATION

Chutisant Kerdvibulvech, Ph.D.

Assistant Professor

ผศ.ดร.ชุตินันต์ เกติวิบูลย์เวช

INSTRUCTOR

Asst. Prof. Dr. Chutisant Kerdvibulvech

ผศ.ดร.ชุตินันต์ เกติวิบูลย์เวช

Graduate School of Communication Arts and Management Innovation
National Institute of Development Administration (NIDA)

- Email: chutisant.ker@nida.ac.th
- Website: www.gscm.nida.ac.th/~chutisant/

2

TODAY'S OVERVIEW

- Instructor's Background
- A Manager 360°: Communication Arts and Management Innovation
 - Technological-based Communication
 - Morning by Asst. Prof. Dr. Chutisant Kerdvibulvech
 - Workshops
 - Afternoon by Assoc. Prof. Dr. Juthamas Keawpijit

Asst. Prof. Dr. Chutisant Kerdvibulvech

3

EDUCATION



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of
Information and Computer Science
Keio University, the oldest university in Japan

Asst. Prof. Dr. Chutisant Kerdvibulvech

4

ACADEMIC WORK EXPERIENCE

- Teaching and being as a research committee in following universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Mahidol University, Kasetsart University
 - NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

Asst. Prof. Dr. Chutisant Kerdvibulvech

5

MEDIA WORK EXPERIENCE (1/3)

1. Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



Asst. Prof. Dr. Chutisant Kerdvibulvech

6

MEDIA WORK EXPERIENCE (2/3)

2. Magazines

- Columnist for Manager 360° Magazine, iTech 360° Magazine, Augmented Reality Magazine



Asst. Prof. Dr. Chutisant Kerdvibulvech

7

MEDIA WORK EXPERIENCE (3/3)

3. Televisions

- IT News Analyst for Nation Channel TV Program



- IT News Analyst for RSU TV Channel



Asst. Prof. Dr. Chutisant Kerdvibulvech

8

TECHNOLOGICAL-BASED COMMUNICATION

DIGITAL MEDIA

Digital media are any media that are encoded in a machine-readable format. [Technology Brief. University of Guelph.]

- Digital Imagery / Digital Video
- Web Pages / Websites
- Digital Audio, e.g. mp3s
- e-books
- Social Media



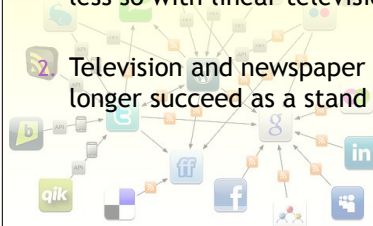
Asst. Prof. Dr. Chutisant Kerdvibulvech

IMPORTANT ROLE OF DIGITAL MEDIA

Online video has become ubiquitous, digital media consumption has become the norm.

Facts about Digital Media

1. Viewers are heavily engaged with digital media - less so with linear television and newspaper.



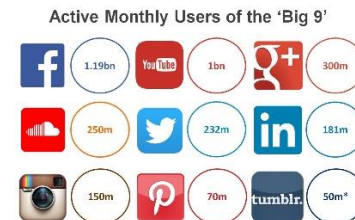
Television and newspaper still matter, but it can no longer succeed as a stand alone platform.

::Source::
Digital Media Trends and the Future of Television, Dr. Seth Geiger, President SmithGeiger LLC (CA, USA)

<https://www.linkedin.com/>

TRENDS FOR DIGITAL MEDIA TRENDS FOR 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
*tumblr does not release active monthly user data and this figure is based on an estimate from Buzzmonitor.com. It is considered that this figure is likely to be around 100m users per month.

kamber

::Source::
Kamber (Sydney, Australia)

kamber | 14 Social Media Trends for 2014

Kamber.com.au | 14

12

MOBILE PHONE AS HOME COMPUTER

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



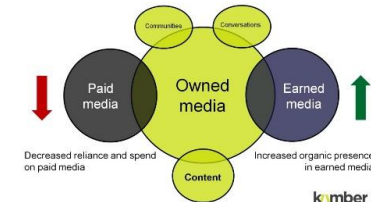
voices.suntimes.com

Asst. Prof. Dr. Chutisant Kerdvibulvech

13

PAID MEDIA / EARNED MEDIA

The content marketing argument



kumber

::Source::
Kamber (Sydney,
Australia)

kumber | 14 Social Media Trends for 2014 | Kamber.com.au | 10

Asst. Prof. Dr. Chutisant Kerdvibulvech

14

CREATIVE-DIGITAL ECONOMY

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But...
 - Easy to build innovatively
 - Create HUGE profits



<http://www.dailynews.co.th/>

Asst. Prof. Dr. Chutisant Kerdvibulvech

THE FUTURE OF DIGITAL MEDIA

Corporate Image and Reputation Management Using Technologies of the Future

- Big Data / Cloud Computing / SEO (Search Engine Optimization) - SEM (Search Engine Marketing)
- Mobile-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)
- New Digital Media



www.3dsq.com.au

Asst. Prof. Dr. Chutisant Kerdvibulvech



“The Oculus community is leading us into the future, and we can’t wait for everyone to experience the magic of VR and the next platform.”...Mark Zuckerberg, CEO of Facebook

24 September 2015

Assistant Professor Dr. Chutisant Kerdvibulvech

3D DISPLAY / 3D TELEVISION

3D DISPLAY / 3D TELEVISION

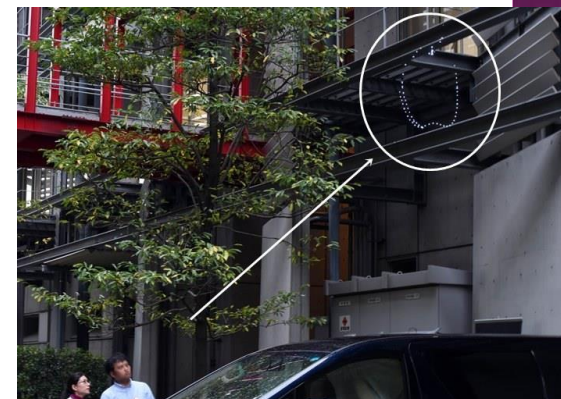


Asst. Prof. Dr. Chutisant Kerdvibulvech

19

3D DISPLAY / 3D TELEVISION

A free-floating image created by firing lasers into thin air



<http://www.afp.com/>

20

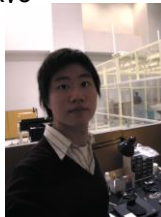
3D DISPLAY / 3D TELEVISION

2014 October

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo

2009

Asst. Prof. Dr. Chutisant Kerdvibulvech



<http://www.afp.com/>

3D DISPLAY / 3D TELEVISION



<http://www.afp.com/>

22

3D DISPLAY / 3D TELEVISION

- Miraikan is located in Odaiba, Tokyo



www.whitehouse.gov



www.japan-guide.com

23

NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,
Carnegie Mellon University (CMU),
United States

SUPER BOWL

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

Asst. Prof. Dr. Chutisant Kerdvibulvech

25

SUPER BOWL

- ◉ “Eye Vision”
- ◉ Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- ◉ The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



Eye Vision from Prof.
Takeo Kanade, CMU

26

THE MATRIX



- ◉ “Virtualized Reality” www.imdb.com
- ◉ People will be able to customize the perspective from which they watch, for example, from that of a particular object.

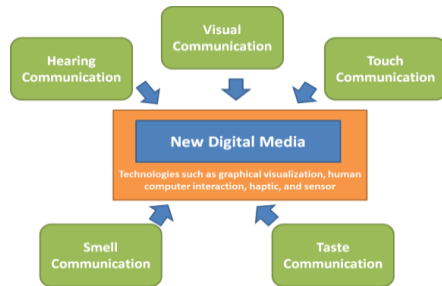


Asst. Prof. Dr. Chutisant Kerdvibulvech

New Viewpoint Analysis 27

NEW DIGITAL MEDIA

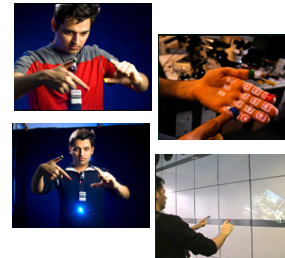
NEW DIGITAL MEDIA



Assistant Professor Dr. Chutisant Kerdvibulvech

VISUAL COMMUNICATION

Augmented Reality



Augmented Reality Magic (card trick), Newmagic Communications Inc.

TED
TALKS

Sixth Sense (MIT)

Assistant Professor Dr. Chutisant Kerdvibulvech

WHAT IS AUGMENTED REALITY (AR)?



- ◉ A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- ◉ Generates a composite view for the user.

Asst. Prof. Dr. Chutisant Kerdvibulvech

31

AUGMENTED REALITY VS. VIRTUAL REALITY

Augmented Reality

- ◉ System augments the real world scene
- ◉ User maintains a sense of presence in real world
- ◉ Needs a mechanism to combine virtual and real worlds

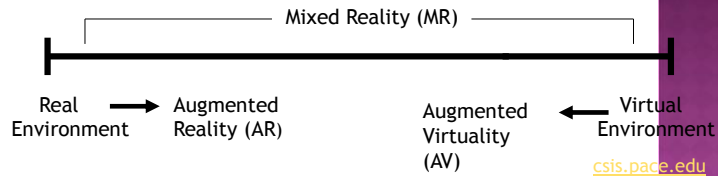
Virtual Reality:

- ◉ Totally immersive environment
- ◉ Visual senses are under control of system

Asst. Prof. Dr. Chutisant Kerdvibulvech

32

MILIGRAM'S REALITY-VIRTUALITY CONTINUUM



Miligram coined the term “Augmented Virtuality” to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

Asst. Prof. Dr. Chutisant Kerdvibulvech

33

COMBINING THE REAL AND VIRTUAL WORLDS

We need:

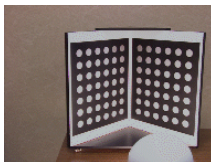
- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

Asst. Prof. Dr. Chutisant Kerdvibulvech

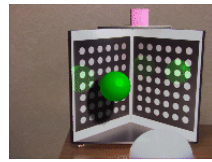
34

COMBINING THE REAL AND VIRTUAL WORLDS (CONT)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



real world



real world with virtual objects and inter-reflections and virtual shading

35

MIXED REALITY

- Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing



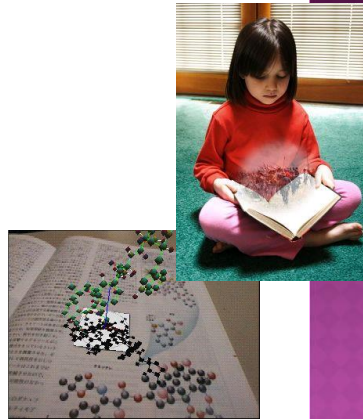
Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

36

Asst. Prof. Dr. Chutisant Kerdvibulvech

This image depicts and artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sony CSL.co.jp

37

Asst. Prof. Dr. Chutisant Kerdvibulvech

RESEARCH ACTIVITIES

- ◉ Develop methods to register the two distinct sets of images and keep them registered in real-time
 - New work in this area has started to use Computer Vision (CV) techniques
- ◉ Develop new display technologies for merging the two images

Asst. Prof. Dr. Chutisant Kerdvibulvech

38

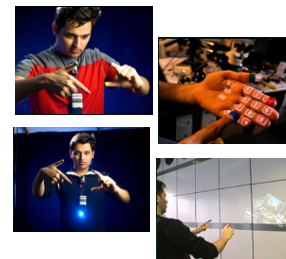
SIXTH SENSE

- ◉ A wearable gestural interface that augments the physical world around us with digital information
- ◉ Let us use natural hand gestures to interact with that information
- ◉ The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Asst. Prof. Dr. Chutisant Kerdvibulvech

39

SIXTH SENSE IN 21ST CENTURY



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED



TED
TALKS

Asst. Prof. Dr. Chutisant Kerdvibulvech

40

AUGMENTED REALITY GUITAR



Dr. Chutisant Kerdvibulvech,
Keio University, Japan
Broadcast in Discovery Channel



Asst. Prof. Dr. Chutisant Kerdvibulvech

41

MIXED REALITY FOR MILITARY APPLICATION



Dr. Adrian Cheok,
National University of Singapore (NUS)
Broadcast in CNN



Assistant Professor Dr. Chutisant Kerdvibulvech

GSCM Digital Campaign (AR)



Pixlive Maker

Pixlive Player

PIXLIVE
MAKER

<http://www.vidinoti.com/>

Assistant Professor Dr. Chutisant Kerdvibulvech



Next Poster



Assistant Professor Dr. Chutisant Kerdvibulvech

NIDA VIRTUALBOX



Assistant Professor Dr. Chutisant Kerdvibulvech

TOUCH COMMUNICATION

Huggable Robot (MIT Media Lab)



<http://robotic.media.mit.edu/portfolio/huggable/>

A blue fancy robotic bear was designed to relieve pain for kids in the Wired Magazine

Assistant Professor Dr. Chutisant Kerdvibulvech

TOUCH COMMUNICATION

Huggy Pajama (Keio)



<http://www.sciencechannel.com/tv-shows/pop-scis-future-of/videos/popscis-future-of-huggy-pajamas/>

Assistant Professor Dr. Chutisant Kerdvibulvech

SMELL COMMUNICATION

A smell-based system called Dollars & Scents inside the entrance to their laboratory at MIT

- The system releases scents and perfumes into the air differently depending on stock market changes.
- Emit a lemon-like fragrance if NASDAQ index is going down, while they emit a rose-like fragrance if the stock market is going up.
- Similarly, they also built an olfaction system, called inStink, for conveying ambient presence and activity awareness for food.

Assistant Professor Dr. Chutisant Kerdvibulvech

KISSENGER

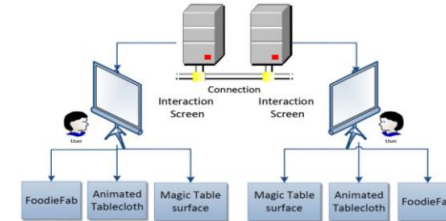
- ◉ Kissenger (National Taipei University)
- ◉ Kiss Messenger
- ◉ Kiss Transmission Robot



Dr. Hooman Samani,
College of Electrical
Engineering and Computer
Science, NTU

Assistant Professor Dr. Chutsant Kerdvibulvech

TASTE COMMUNICATION



The Food Media's system configuration is set up by
[Wei13a] from National University of Singapore.

Assistant Professor Dr. Chutsant Kerdvibulvech

FOOD MEDIA

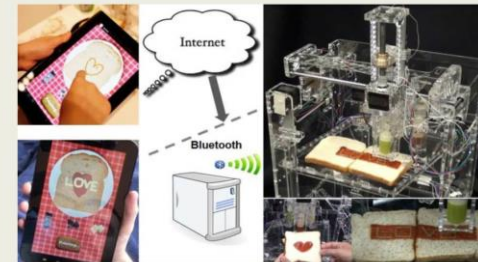


WEI JUN, Food Media: Food and Technology as a Medium for Social
Communication, PhD Dissertation from National University of Singapore, 2013

Assistant Professor Dr. Chutsant Kerdvibulvech

FOOD MEDIA

Food Messaging



WEI JUN, Food Media: Food and Technology as a Medium for Social Communication,
PhD Dissertation from National University of Singapore, 2013

Assistant Professor Dr. Chutsant Kerdvibulvech

Food Media

- ◉ To utilize interactive activities to reconstruct the missing physical and multi-sensory experience of food in remote dining
- ◉ Core concept:
 - To create ways for social communication with interactive and emotional experience supported by food and technology

Assistant Professor Dr. Chutisant Kerdvibulvech

ELECTRIC TASTE MACHINE



N. Ranasinghe et al, Simulating the sensation of taste for immersive experiences, 2013 ACM international workshop on Immersive media experiences

Assistant Professor Dr. Chutisant Kerdvibulvech

DIFFICULTY FOR TASTE AND SMELL

- ◉ How to digitize taste and smell..
- ◉ Issue of Chemistry
- ◉ Issue of Likeness
- ◉ Issue of Safety

Assistant Professor Dr. Chutisant Kerdvibulvech

MULTISENSORY HUMAN COMMUNICATION (1)

Adrian David Cheok, Professor of Pervasive Computing at City University London
Young Global Leader 2008 by the World Economic Forum



<http://www.weforum.org/>

Assistant Professor Dr. Chutisant Kerdvibulvech

MULTISENSORY HUMAN COMMUNICATION (2)



TEDxBBarcelona



<http://tedxtalks.ted.com/>

Assistant Professor Dr. Chutisant Kerdvibulvech

TECHNOLOGY DELIVERS A TASTE OF FUTURE INTERNET COMMUNICATION



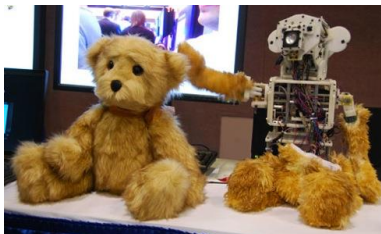
AP

<http://ap.org/>

Assistant Professor Dr. Chutisant Kerdvibulvech

TOUCH COMMUNICATION

Huggable Robot (MIT)



Wired Magazine

<http://robotic.media.mit.edu/portfolio/huggable/>

Assistant Professor Dr. Chutisant Kerdvibulvech

QUESTIONS?

Asst. Prof. Dr. Chutisant Kerdvibulvech
Graduate School of Communication Arts and Management Innovation,
National Institute of Development Administration (NIDA)
chutisant.ker@nida.ac.th
<https://www.facebook.com/chutisant.kerdvibulvech>

Assistant Professor Dr. Chutisant Kerdvibulvech